

A Post-Quantum Relation-Bound Blind Certification Core with Signature-Gated Threshold Decryption

Research construction draft, version 1.6

23 August 2026

Abstract

This document specifies and proves an abstract post-quantum cryptographic core for privacy-preserving, accountable credentials. During an offline issuance phase, a holder obtains a pair of blind signatures only after proving that the hidden ticket, the blind-signing request, the holder secret, and a trace ciphertext are mutually consistent. During opening, threshold decryption shares are released only for a validly signed ticket and an independently authorized case. This *signature-gated threshold decryption* rule replaces a public validity proof carried by every online ticket.

The main result is conditional and modular: assuming simulation-extractable zero knowledge, Blind-UOV one-more unforgeability and blindness, and a compact threshold encryption scheme with post-quantum privacy and unique openings, the construction provides issuer unlinkability, certified ciphertext validity, decoder safety, trace correctness, non-frameability, and a gated chosen-ciphertext privacy notion against fewer than the threshold number of opening authorities. The result deliberately does not claim ordinary IND-CCA security for the bare trace encryption. Version 1.2 also instantiates the trace relation as a Niederreiter syndrome plus a SHAKE256/KMAC256 one-time hash DEM, proves information-theoretic uniqueness of the decoded error, and removes a redundant ticket commitment. Version 1.3 freezes a paper-compatible Blind-UOV-Is/Shorter costing profile, fixes a 368-byte payload, gives a reproducible five-block circuit count, and separates the large issuance NIZK from the CAP proof carried by the final Blind-UOV signature. Version 1.4 adds an executable witness builder, a streaming characteristic-two circuit generator, independent SHAKE256/KMAC256 checks, and full-circuit rejection tests. Native Blind-UOV circuit integration, backend simulation extractability, and the threshold implementation remain explicit obligations. Version 1.5 lowers the complete incremental relation into a checksummed binary $\mathbb{F}_2$ -R1CS archive, independently parses every row, and separates the nonlinear-only cost model from the portable materialized-row count. Version 1.6 corrects the signer-view ABI to the exact Blind-UOV protocol: the signer sees only y , while the CAP commitment, message, mask, and commitment randomness remain hidden until finalization. It also identifies a cross-message claw obligation absent from ordinary blind-signature security. The aggressive profile therefore uses two independent, domain-separated Blind-UOV-Is lanes. Its online ticket is 7,912 bytes; its security theorem is conditional on a QROM claw-resistance reduction and native circuit integration.

1 Status, scope, and non-goals

This is the standalone cryptographic core. It intentionally excludes satellite AKE, radio transport, FLEO relay behavior, handover, revocation distribution, and concrete parameter optimization. Issuance and opening may use large proofs, large public keys, and large private keys. The constrained online path carries only a ticket payload and two compact blind signatures.

The construction is designed for the following division of trust.

- The issuing HNCC is *honest but curious*: it authenticates the real identity, enforces a common issuance policy, verifies Π_{issue} , and signs only after acceptance, but later tries to link an observed ticket to an issuance session.

- Holders and network attackers may be malicious and quantum polynomial time.
- Fewer than τ of n opening authorities may be corrupted. Any authorized set of at least τ authorities can open a ticket.
- Issuer signing keys and opening decryption keys are independent even when the organizations operating them overlap.
- Per-user visible metadata is forbidden. Version, epoch, policy, expiration bucket, and key identifiers are common to a sufficiently large anonymity set; otherwise they become an issuer watermark outside the cryptographic game.

Claim boundary. The theorems below are complete reductions for the abstract interfaces and games defined here. They are not a claim that the present Blind-UOV or threshold Niederreiter papers already implement every required interface. The Blind-UOV request equations and their visibility are fixed in Section 3.2; the exact added circuit is counted in Section 8.3 and executed in Section 8.4. The resulting proof-size and prover runtime figures remain paper-anchored estimates rather than implementation benchmarks. The executable generator is a relation validator, not a proof backend. The numerical Niederreiter set is a size anchor, not a deployment approval.

2 Notation and post-quantum conventions

The security parameter is λ . All adversaries and reductions are uniform or non-uniform QPT algorithms. A function $\text{negl}(\lambda)$ is negligible. Encodings are canonical, length-prefixed, and domain separated. Concatenation is written $\parallel$. Every hash invocation includes a fixed protocol label and version.

The issuing organization authenticates a real identity rid before the cryptographic issuance protocol. The session identifier sid is used only for replay and quota control and is never copied into a ticket. The common issuance information is

$$\text{ci} = (\text{version}, \text{epoch}, \text{domain}, \text{policy}, \text{expiryBucket}, \text{kid}_{\text{OA}}, \text{kid}_{\text{I}}).$$

Its fixed 32-byte ticket representation is

$$\text{ctx} = \text{SHAKE256}(\text{PQ-RBBC}/\text{CTX} \parallel \text{Encode}(\text{ci}), 256).$$

The federation configuration signature authenticates (ci, ctx) . The digest is computed once outside Π_{issue} and is a public input; the issuer cannot select a per-user value.

For any game G , $\text{Adv}_{\mathcal{A}}^G(\lambda)$ denotes the usual absolute distance from the ideal winning probability. Oracle calls are polynomially bounded.

3 Required primitive interfaces

3.1 Hashing and CAP commitment

We use domain-separated hashes

$$H_{\text{ticket}}, \quad H_{\text{BUOV}}, \quad H_{\text{hold}}, \quad H_{\text{evidence}}.$$

They are modeled as quantum-resistant collision-resistant hashes. A random oracle or QROM instantiation may be used by a concrete component, but the core proof calls only the explicitly named property.

For the version-1.3 circuit audit, fix

$$\begin{aligned} H_{\text{ticket}}(X) &= \text{SHAKE256}(\text{PQ-RBBC}/\text{TICKET} \parallel X, 256), \\ H_{\text{hold}}(k) &= \text{SHAKE256}(\text{PQ-RBBC}/\text{HOLD} \parallel k, 256). \end{aligned}$$

The Blind-UOV hash H_{BUOV} and CAP commitment internals use the paper’s Anemoi-based costing profile; they are already included in its baseline NIZK count. This mixed choice retains standardized SHAKE/KMAC for new protocol logic while avoiding the multi-billion-constraint SHA3+AES baseline inside the Blind-UOV commitment proof.

There is no separate public commitment to the ticket payload. The accepting proof extracts M , Equation (I2) fixes its digest, and the Blind-UOV request binds that digest. Publishing an additional c_M would add a primitive and a circuit block without strengthening this chain.

3.2 Blind-UOV request normal form and visibility

The blind-signature interface is

$$(\text{bsk}, \text{bvk}) \leftarrow \text{BUOV.KeyGen}(1^\lambda),$$

with an interactive request/sign/unblind protocol and a public verification algorithm

$$\text{BUOV.Vfy}(\text{bvk}, m, \sigma).$$

The core needs more than ordinary blindness: the issuance proof must identify the exact hidden message represented by the request. Protocols 3 and 4 of the 31 October 2025 revision have the user compute [1]

$$c_r \leftarrow \text{CAP.Commit}(r; \rho), \tag{B1}$$

$$y = r + H_{\text{BUOV}}(m, c_r) \in \mathbb{F}_q^o. \tag{B2}$$

Only (y, π_1) is sent to the signer. The values m, r, ρ, c_r are witness/state, not signer-visible request fields. The final signature later contains (c_r, c_x, π_2) . Publishing c_r during issuance would both depart from the paper and give the issuer a direct link to the final signature.

A single Blind-UOV-Is target is 256 bits. Our authorization-binding property is a freely chosen cross-message collision/claw problem, not a fixed-target preimage problem. Generic quantum collision algorithms therefore give only about $2^{256/3} = 2^{85}$ work [2, 3]. The aggressive post-quantum profile runs two independently randomized and domain-separated lanes $i \in \{0, 1\}$. Define $\mu_i = \text{Encode}(\text{PQ-RBBC}/\text{BUOV-LANE}, i, m)$ and compute:

$$c_{r,i} \leftarrow \text{CAP.Commit}_i(r_i; \rho_i), \tag{B3}$$

$$y_i = J_m^i(r_i, \rho_i) = r_i + H_{\text{BUOV}}(\mu_i, c_{r,i}). \tag{B4}$$

The public request is exactly $\beta = (y_0, y_1)$; its hidden witness is $b = ((r_0, \rho_0), (r_1, \rho_1))$. Masks and commitment randomness MUST be independent across lanes. The issuer returns two ordinary Blind-UOV responses, and finalization produces ordinary signatures on μ_0 and μ_1 . They bind one logical ticket digest m but are not byte-identical messages.

Definition 3.1 (Dual Blind-UOV request relation). *For $\beta = (y_0, y_1)$ and $b = ((r_0, \rho_0), (r_1, \rho_1))$, define*

$$R_{\text{req}}^{\text{BUOV},2}(\beta; m, b) = 1 \iff \bigwedge_{i=0}^1 [c_{r,i} = \text{CAP.Commit}_i(r_i; \rho_i) \wedge y_i = r_i + H_{\text{BUOV}}(\mu_i, c_{r,i})].$$

The $c_{r,i}$ are derived hidden witnesses. A completed session returns σ_0, σ_1 , where σ_i verifies on the canonical μ_i derived from the same m .

Assumption 3.2 (Dual-lane cross-message request claw resistance). *No QPT adversary can output $m \neq m'$ and valid independent openings for both sides such that*

$$(J_m^0(r_0, \rho_0), J_m^1(r_1, \rho_1)) = (J_{m'}^0(r'_0, \rho'_0), J_{m'}^1(r'_1, \rho'_1))$$

except with negligible probability. For ideal independent 256-bit lane outputs, the generic quantum collision scale is about $2^{512/3} = 2^{170}$. This estimate motivates the construction but is not a completed QROM reduction for the structured J^i maps.

We additionally require correctness, parallel two-instance blindness, and one-more unforgeability of Blind-UOV. Assumption 3.2 is a new composition obligation; it is not implied merely by CAP binding and ordinary collision resistance of H_{BUOV} .

Remark 3.3 (Why ordinary blind signing is insufficient). *If Π_{issue} proves a valid ciphertext but the proof is not tied to the exact blind request β , a malicious holder can prove statement M_0 and ask for a signature on M_1 . The $R_{\text{req}}^{\text{BUOV},2}$ conjunct is the formal connection that prevents this substitution.*

3.3 Simulation-extractable NIZK argument of knowledge

The proof system $\Pi_{\text{issue}} = (\text{Setup}, \text{Prove}, \text{Vfy})$ is a post-quantum simulation-extractable NIZK argument of knowledge for the issuance relation in Section 4. Its setup produces a CRS; the security proof may use simulation and extraction trapdoors. In the real protocol no trapdoor is available to any party.

The proof may be several megabytes because it is transmitted only during infrequent issuance. This design choice is intentional. Knowledge soundness alone is enough for decoder safety in a simple stand-alone execution; simulation extractability is used to support concurrent sessions and the gated-CCA hybrid proof.

3.4 Compact threshold trace encryption

The threshold encryption interface is

$$\begin{aligned} (\text{tpk}, \text{tsk}_1, \dots, \text{tsk}_n) &\leftarrow \text{TEnc.KeyGen}(1^\lambda, n, \tau), \\ C &\leftarrow \text{TEnc.Enc}(\text{tpk}, \mu; \text{ad}, \rho), \\ d_i &\leftarrow \text{TPDec}(\text{tsk}_i, C), \\ \mu \text{ or } \perp &\leftarrow \text{TComb}(C, \{d_i\}_{i \in I}), \quad |I| \geq \tau. \end{aligned}$$

There is an NP relation

$$R_{\text{enc}}(\text{tpk}, C, \text{ad}; \mu, \rho) = 1 \iff C = \text{TEnc.Enc}(\text{tpk}, \mu; \text{ad}, \rho).$$

Only ciphertexts in the language L_{enc} are promised to be safe inputs to the distributed decoder.

We require the following post-quantum properties.

- (T1) **Correctness on valid inputs.** Combining any τ correct shares for a ciphertext in L_{enc} returns its plaintext.
- (T2) **τ -privacy under chosen plaintext.** A QPT adversary holding fewer than τ secret shares cannot distinguish encryptions of two equal-length plaintexts.
- (T3) **Unique opening.** It is infeasible to find $C, \mu, \rho, \mu', \rho'$ with $\mu \neq \mu'$ such that both $R_{\text{enc}}(\text{tpk}, C, \text{ad}; \mu, \rho) = 1$ and $R_{\text{enc}}(\text{tpk}, C, \text{ad}; \mu', \rho') = 1$.
- (T4) **Robust shares.** Invalid or inconsistent partial decryptions are detected by TComb or by a component proof/MAC before a plaintext is accepted.

Ordinary IND-CCA security of the bare encryption is not required and is not claimed.

3.5 Opening authorization

An independent post-quantum authorization mechanism verifies

$$\text{GateAuthVfy}(\text{gavk}, d_M, \text{caselD}, H_{\text{evidence}}(E), \text{auth}) = 1,$$

where $d_M = H_{\text{ticket}}(\text{Encode}(M))$. The authorization binds a ticket digest, case identifier, evidence digest, expiry, and purpose. Its unforgeability and policy governance are orthogonal to trace-ciphertext privacy, but the binding is required to prevent using an authorization for a different ticket.

4 Ticket format and issuance relation

The hidden ticket payload is

$$M = (\text{ctx}, \text{sn}, h, C), \quad h = H_{\text{hold}}(k_{\text{hold}}), \quad C = \text{Ntr.Enc}(\text{tpk}, \text{ctx}, h; \text{rid}, \text{sn}, e).$$

For the v1.3 profile all fields are fixed width and

$$\text{Encode}(M) = \text{ctx} \parallel \text{sn} \parallel h \parallel s \parallel c_\mu \parallel \gamma \in \{0, 1\}^{2944},$$

namely $32 + 16 + 32 + 208 + 48 + 32 = 368$ bytes. No variable-length field or user-selected policy string occurs in the payload. The serial number $\text{sn} \leftarrow \{0, 1\}^{128}$ and holder key $k_{\text{hold}} \leftarrow \{0, 1\}^{256}$ are fresh. The serial appears both outside and inside the trace ciphertext so that an opener can detect payload splicing. The holder secret itself never appears in the ticket. Define

$$m = H_{\text{ticket}}(\text{Encode}(M)).$$

The public issuance statement and private witness are

$$\begin{aligned} x &= (\text{pp}, \text{ctx}, \text{sid}, \text{rid}, \beta), \\ w &= (M, (r_0, \rho_0), (r_1, \rho_1), k_{\text{hold}}, e). \end{aligned}$$

The relation $R_{\text{issue}}(x; w) = 1$ iff all of the following hold:

$$\text{TicketShape}(M, \text{ctx}) = 1, \tag{I1}$$

$$m = H_{\text{ticket}}(\text{Encode}(M)), \tag{I2}$$

$$R_{\text{req}}^{\text{BUOV}, 2}(\beta; m, ((r_0, \rho_0), (r_1, \rho_1))) = 1, \tag{I3}$$

$$M.h = H_{\text{hold}}(k_{\text{hold}}), \tag{I4}$$

$$R_{\text{enc}}(\text{tpk}, M.C, \text{Encode}(\text{ctx}, M.\text{sn}, M.h); \text{rid} \parallel M.\text{sn}, e) = 1. \tag{I5}$$

The real identity and session identifier are public to the issuer during issuance but are not part of M . Equation (I5) binds the hidden trace plaintext to the identity authenticated in that exact session.

5 Construction

5.1 Setup

$\text{Core.Setup}(1^\lambda, n, \tau)$ performs:

1. generate Π_{issue} parameters and the domain-separated hash parameters;
2. generate issuer keys $(\text{bsk}, \text{bvk}) \leftarrow \text{BUOV.KeyGen}(1^\lambda)$;
3. generate independent opening keys $(\text{tpk}, \text{tsk}_1, \dots, \text{tsk}_n) \leftarrow \text{TEnc.KeyGen}(1^\lambda, n, \tau)$;
4. publish (ci, ctx) authenticated by a federation configuration signature.

No party is allowed to derive bsk and any tsk_i from the same seed or key hierarchy.

5.2 Blind relation-bound issuance

For authenticated identity rid and fresh issuer-side sid :

1. The holder samples $\text{sn}, k_{\text{hold}}, e$ and computes

$$h = H_{\text{hold}}(k_{\text{hold}}), \quad C = \text{Ntr.Enc}(\text{tpk}, \text{ctx}, h; \text{rid}, \text{sn}, e)$$

according to Equations (N1)–(N7), constructs M , and computes $m = H_{\text{ticket}}(\text{Encode}(M))$.

2. For $i \in \{0, 1\}$ it independently samples r_i, ρ_i , computes

$$\begin{aligned} \mu_i &= \text{Encode}(\text{PQ-RBBC/BUOV-LANE}, i, m), \\ c_{r,i} &= \text{CAP.Commit}_i(r_i; \rho_i), & y_i &= r_i + H_{\text{BUOV}}(\mu_i, c_{r,i}). \end{aligned}$$

sets the public request $\beta = (y_0, y_1)$, and produces

$$\pi_{\text{issue}} \leftarrow \Pi_{\text{issue}}.\text{Prove}(\text{crs}, x, w).$$

3. The issuer checks authenticated identity, epoch, quota, replay status, and

$$\Pi_{\text{issue}}.\text{Vfy}(\text{crs}, x, \pi_{\text{issue}}) = 1.$$

Only then does it execute, for both lanes, $z_i \leftarrow \text{BUOV.Respond}(\text{bsk}, y_i)$.

4. For both lanes the holder computes

$$\sigma_i \leftarrow \text{BUOV.Finalize}(\mu_i, c_{r,i}, y_i, z_i; r_i, \rho_i)$$

and accepts only if both $\text{BUOV.Vfy}(\text{bvk}, \mu_i, \sigma_i) = 1$.

The proof π_{issue} is not an extra proof placed beside the Blind-UOV well-built-request proof. It is the strengthened replacement for that proof: the original request equations are retained and the ticket, identity, holder, and trace-ciphertext conjuncts are appended to the proved relation. Thus issuance sends one large relation proof, not two independent proofs.

The spendable ticket is $T = (M, \sigma_0, \sigma_1)$. Neither β , π_{issue} , rid , nor sid is transmitted with T .

5.3 Ticket verification

$\text{Core.VerifyTicket}(T)$ recomputes $m = H_{\text{ticket}}(\text{Encode}(M))$ and both canonical μ_i encodings, checks canonical payload encoding and the public common-information policy, then returns one iff

$$\text{BUOV.Vfy}(\text{bvk}, \mu_0, \sigma_0) = \text{BUOV.Vfy}(\text{bvk}, \mu_1, \sigma_1) = 1.$$

This check proves neither public ciphertext validity nor identity correctness by itself. Those facts follow from the certified-issuance theorem and the trust assumption that the issuer signs only after accepting Π_{issue} .

5.4 Signature-gated threshold opening

For an opening request

$$Q = (T, E, \text{caselD}, \text{auth}),$$

each honest opening authority runs the following gate before touching its decoder:

1. require $\text{Core.VerifyTicket}(T) = 1$;

2. require the common epoch and key identifiers to be acceptable;
3. compute $d_M = H_{\text{ticket}}(\text{Encode}(M))$ and require a valid, unused, purpose-limited authorization bound to $(d_M, \text{caseID}, H_{\text{evidence}}(E))$;
4. only after all checks pass, output $d_i = \text{TPDec}(\text{tsk}_i, M.C)$, bound to the same request digest.

The combiner accepts a reconstructed plaintext $\text{rid}' \parallel \text{sn}'$ only if robust share verification succeeds and $\text{sn}' = M.\text{sn}$. Otherwise it outputs $\perp$.

Critical API rule. The implementation MUST NOT export $\text{TPDec}(\text{tsk}_i, C)$ as a callable production endpoint. The only exposed operation is $\text{Core.OpenShare}(\text{tsk}_i, Q)$, which internally performs the gate. This is a security boundary, not an application convention.

6 Security games

6.1 Certified-language soundness

The challenger maintains a log $\mathcal{L}$ of accepted issuance sessions. In the analysis it uses the extractor to associate each completed session with (rid, M, w) . An adversary wins CertSound if it outputs a ticket $T = (M, \sigma_0, \sigma_1)$ for which $\text{Core.VerifyTicket}(T) = 1$ and either:

- (a) M is not associated with any completed accepted issuance in $\mathcal{L}$; or
- (b) every associated extraction fails one of Equations (I1)–(I5).

Distinct messages are counted by their canonical ticket digest. This game captures both a new signature and a proof/request substitution.

6.2 Issuer unlinkability

An honest-protocol curious issuer chooses or observes two authenticated identities $\text{rid}_0, \text{rid}_1$ that receive the same ci. Two honest holders execute issuance concurrently. The issuer obtains its full protocol views, after which it receives the two resulting tickets in random order and guesses the order. It may retain all issuance state but receives no opening shares. The advantage over $1/2$ is issuer-unlinkability advantage.

This game allows the issuer to know which identity participated in each issuance; it hides which later ticket came from which session. It cannot hide timing, a uniquely sized policy, or a per-user expiration bucket, which is why common metadata is a stated system requirement.

6.3 Gated chosen-ciphertext trace privacy

The game t-GCCA is parameterized by threshold τ . The adversary receives all public parameters and fewer than τ opening secret shares. It may run malicious-user issuance sessions for identities it is authorized to use, receiving completed tickets, and may submit opening queries for any validly authorized ticket other than the exact challenge ticket digest.

For the challenge it selects equal-length identities $\text{rid}_0, \text{rid}_1$ and common ticket metadata. The challenger samples $b \leftarrow \{0, 1\}$, forms a valid ticket whose trace ciphertext encrypts $\text{rid}_b \parallel \text{sn}^*$, and returns $T^* = (M^*, \sigma^*)$. The adversary wins by guessing b .

The HNCC issuer is not corrupted in this game because it necessarily observes the authenticated identity at issuance. HNCC privacy is instead covered by issuer unlinkability. The game models an outside observer, an evidence processor, and fewer than τ opening authorities.

Remark 6.1 (Why the game is not ordinary CCA). *The opening oracle accepts only issuer-certified ticket payloads, not arbitrary ciphertexts. Calling the result IND-CCA for bare TEnc would be incorrect. The security statement is protocol-specific: certification plus the gate turns chosen-ciphertext access into chosen-certified-ciphertext access.*

6.4 Trace soundness and non-frameability

An adversary wins TraceSound if a valid ticket produced from a completed accepted issuance authenticated as rid opens to $\text{rid}' \neq \text{rid}$, or if it opens with an external serial number different from the encrypted serial. The issuer is honest in following the protocol. The holder may be malicious.

7 Security results

7.1 Correctness

Theorem 7.1 (End-to-end correctness). *For an honest issuance authenticated as rid , an honestly produced ticket $T = (M, \sigma_0, \sigma_1)$ verifies. Any authorized set of at least τ honest opening authorities reconstructs $\text{rid} \parallel M.\text{sn}$, except with the sum of the component correctness errors.*

Proof. Honest computation satisfies every conjunct of R_{issue} , so Π_{issue} accepts. Blind-signature correctness gives two verifying signatures on the two lane encodings of $H_{\text{ticket}}(\text{Encode}(M))$. Equation (I5) gives a valid encryption of $\text{rid} \parallel M.\text{sn}$. Threshold-encryption correctness on valid inputs reconstructs that plaintext from any τ correct shares. The serial cross-check therefore succeeds. $\square$

7.2 Blind-signing message binding

Lemma 7.2 (Conditional dual-request uniqueness). *Let $\beta = (y_0, y_1)$. If the dual request relation accepts valid openings for both m and $m' \neq m$, then those values constitute a winning output in the game of Assumption 3.2.*

Proof. Expanding the two accepted relations gives two independently opened pairs of J values equal to the same public (y_0, y_1) for distinct messages. This is exactly the dual cross-message claw game. Notice that CAP binding alone cannot rule it out because the two sides may use different commitments. $\square$

Lemma 7.3 (Request/proof binding). *Suppose an issuance proof for β is accepting and extractable. Except with dual-request claw probability, the message signed through β is exactly $H_{\text{ticket}}(\text{Encode}(M))$ for the payload M in the extracted witness. A different payload cannot verify under the same signed digest except by finding a collision in H_{ticket} .*

Proof. Extraction yields $(M, (r_0, \rho_0), (r_1, \rho_1), \dots)$ satisfying Equations (I2)–(I3). Equation (I2) fixes m to the canonical ticket digest. Lemma 7.2 makes the message represented by β equal to this m . If a different canonical payload $M' \neq M$ verifies under that signed value, then $H_{\text{ticket}}(\text{Encode}(M')) = H_{\text{ticket}}(\text{Encode}(M))$, a collision. $\square$

Theorem 7.4 (Certified-language soundness). *For any QPT adversary $\mathcal{A}$, there exist QPT reductions $\mathcal{B}_1, \dots, \mathcal{B}_4$ such that*

$$\text{Adv}_{\text{Core}, \mathcal{A}}^{\text{CertSound}} \leq \text{Adv}_{\Pi_{\text{issue}}, \mathcal{B}_1}^{\text{SE-AoK}} + 2\text{Adv}_{\text{BUOV}, \mathcal{B}_2}^{1\text{mUF}} + \text{Adv}_{J, \mathcal{B}_3}^{\text{dualClaw}} + \text{Adv}_{H_{\text{ticket}}, \mathcal{B}_4}^{\text{coll}}.$$

Consequently, every verifying ticket is associated with an accepted issuance witness satisfying R_{issue} , except with negligible probability.

Proof. Consider a verifying output (M, σ_0, σ_1) . If its canonical digest was not the message of a completed blind-signing session, then the output is a one-more forgery. Otherwise consider the corresponding accepting issuance proof. Simulation extraction returns a witness or violates the SE-AoK property. Lemma 7.3 associates the signed digest with the extracted payload, except under the remaining claw and collision bad events. The extracted witness then satisfies every issuance conjunct, including ciphertext validity and identity binding. $\square$

Corollary 7.5 (Certified ciphertext validity). *If $\text{Core.VerifyTicket}(M, \sigma_0, \sigma_1) = 1$, then $M.C \in L_{\text{enc}}$ and it has an opening to the authenticated issuance identity and the external serial number, except with the bound in Theorem 7.4.*

7.3 Decoder safety

Theorem 7.6 (Signature-gated decoder safety). *Assume honest opening authorities expose only Core.OpenShare and execute its checks in the specified order. The probability that an honest authority evaluates TPDec on a ciphertext outside L_{enc} is at most*

$$\text{Adv}_{\text{Core}}^{\text{CertSound}} + \text{Adv}_{\text{GateAuth}}^{\text{forge}} + \text{Adv}_{H_{\text{ticket}}}^{\text{coll}}.$$

Proof. Reaching TPDec implies that the ticket signature and the opening authorization both verified for the same ticket digest. By Corollary 7.5, a verifying ticket contains a ciphertext in L_{enc} . A mismatch between the ticket checked by the gate and the ciphertext passed to the decoder either violates the atomic API invariant or requires a digest collision. A request not authorized for that digest is an authorization forgery. Taking a union bound gives the result. $\square$

Remark 7.7. *This theorem explains what is gained by moving the validity proof out of the online ticket. The large issuance proof is amortized, while the small issuer signature becomes a certificate that the ciphertext was once proved valid. The theorem fails immediately if an implementation offers a raw partial- decryption endpoint.*

7.4 Issuer unlinkability

Theorem 7.8 (Issuer unlinkability). *For an honest-protocol curious issuer, common visible metadata, and two honest holders, there exist QPT reductions such that*

$$\text{Adv}_{\text{Core}}^{\text{unlink}} \leq 2\text{Adv}_{\text{BUOV}}^{\text{blind}} + 2\text{Adv}_{\text{II}_{\text{issue}}}^{\text{zk}} + \text{Adv}_{\text{TEnc}}^{\text{t-IND-CPA}} + \text{Adv}_{H_{\text{ticket}}}^{\text{coll}}.$$

Proof. First replace each real issuance proof with a simulated proof; zero knowledge bounds the change. The remaining request/response views are two parallel blind-signature views per hidden message. A two-hybrid argument applies Blind-UOV blindness once per lane and permits swapping those messages without the issuer detecting the swap. Threshold-encryption privacy prevents the issuer, which holds no opening threshold, from matching a final trace ciphertext to either known issuance identity. Canonical digest collisions are excluded. After these replacements, the issuer's view is independent of the final permutation, so its success probability is $1/2$. $\square$

Operational qualifier. The theorem is cryptographic, not traffic-analytic. Issuance batching, fixed message sizes, delayed activation, and common metadata are required to keep timing and formatting from becoming a link tag.

7.5 Trace correctness and non-frameability

Theorem 7.9 (Trace soundness). *Under certified-language soundness and unique opening of TEnc , a malicious holder completing issuance while authenticated as rid cannot produce a verifying ticket that opens to $\text{rid}' \neq \text{rid}$ or to a different serial number, except with negligible probability.*

Proof. By Theorem 7.4, a verifying ticket is associated with an extracted accepted issuance witness. Equation (I5) states that its ciphertext opens to $\text{rid}||M.\text{sn}$. If threshold opening returns a different plaintext while all robust-share checks pass, either threshold correctness fails or the ciphertext has two different valid openings. The latter violates unique opening. The explicit serial comparison catches any external/internal serial mismatch. $\square$

Remark 7.10 (Issuer framing). *The theorem assumes an honest-protocol issuer because the issuer learns rid and controls the signing key. A malicious issuer could directly sign a payload encrypting an arbitrary victim identity. Removing that trust requires a threshold blind issuer, an auditable issuance log with privacy-preserving accountability, or retaining a publicly checked proof. It is not solved by the present compact-ticket core.*

7.6 Gated chosen-ciphertext trace privacy

Theorem 7.11 (τ -GCCA trace privacy). *Assume fewer than τ opening authorities are corrupted. For every QPT adversary $\mathcal{A}$ in the game of Section 6.3, there exist QPT reductions $\mathcal{B}_i$ such that*

$$\begin{aligned} \text{Adv}_{\text{Core}, \mathcal{A}}^{\text{t-GCCA}} &\leq \text{Adv}_{\text{TEnc}, \mathcal{B}_0}^{\text{t-IND-CPA}} + \text{Adv}_{\Pi_{\text{issue}}, \mathcal{B}_1}^{\text{SE-AoK}} + 2\text{Adv}_{\text{BUOV}, \mathcal{B}_2}^{\text{1mUF}} \\ &\quad + \text{Adv}_{\text{TEnc}, \mathcal{B}_3}^{\text{unique}} + \text{Adv}_{J, \mathcal{B}_4}^{\text{dualClaw}} \\ &\quad + \text{Adv}_{H_{\text{ticket}}, \mathcal{B}_5}^{\text{coll}}. \end{aligned}$$

Proof. We use a sequence of games.

Game 0. This is the real t-GCCA game.

Game 1: extract accepted issuance sessions. For every accepted adversarial issuance proof, use the SE-AoK extractor to record

$$(d_M, M, \text{rid}, e, k_{\text{hold}}, (r_0, \rho_0), (r_1, \rho_1), \dots)$$

in a private table $\mathcal{L}$. Abort if an accepting proof has no valid witness. The difference from Game 0 is bounded by SE-AoK security. Request binding and hash collision resistance ensure that the final signed digest is the digest recorded in $\mathcal{L}$.

Game 2: reject unlogged verifying tickets. If the adversary submits a verifying ticket whose digest is neither the challenge digest nor in $\mathcal{L}$, abort. Such a ticket supplies an additional valid message/signature pair not obtained from a completed issuance, which yields a one-more forgery. Proof/request substitution and digest ambiguity are charged to the corresponding terms in the theorem.

Game 3: simulate the opening oracle from the issuance table. For every non-challenge verifying ticket, look up the extracted plaintext in $\mathcal{L}$ and return it after the authorization checks. This has the same result as threshold opening by correctness and unique opening, but it uses no honest opening secret share and never invokes a decryption oracle. A ticket that reuses the challenge ciphertext under a newly certified digest requires an accepted proof containing a valid opening of that ciphertext. Extraction then reveals the challenge plaintext. If the extracted opening disagrees with the challenge plaintext, it violates unique opening; if it agrees, the reduction can immediately guess the threshold-encryption challenge bit. Thus challenge re-certification does not create an unanswerable decryption query.

Game 4: replace the challenge encryption. Embed the τ -IND-CPA challenge of TEnc as $M^*.C$. The reduction is given fewer than τ shares exactly as in the base game. It samples the remaining visible ticket fields identically. Because it generated the issuer key and knows the complete challenge message m^* , it locally executes both sides of both genuine Blind-UOV lanes and obtains two exactly distributed final signatures; it does not need the unknown trace-encryption randomness. The challenge issuance transcript is not given to the adversary in this game, and π_{issue} influences only the accept/reject gate, not the distribution of the response after acceptance. Therefore the reduction may internally bypass that gate when manufacturing the challenge ticket without changing any value visible to the adversary. All opening queries are answered from $\mathcal{L}$ as in Game 3. Therefore changing the encrypted identity from rid_0 to rid_1 changes the adversary’s view by at most the τ -IND-CPA advantage of TEnc.

In the final game the challenge bit is independent of the adversary’s view. Summing the hybrid differences proves the theorem. $\square$

Why the reduction is useful. The large issuance proof is not merely a validity check. Its extractor gives the simulator the plaintext and randomness of every non-challenge ciphertext that can ever obtain a valid issuer signature. Therefore the simulator can answer gated opening queries without a base decryption oracle. This is the precise reason a CPA-secure compact threshold encryption can suffice inside the gated protocol even though it would not suffice under an unrestricted CCA API.

7.7 One-more ticket unforgeability

Theorem 7.12 (Ticket one-more unforgeability). *After q completed accepted dual-lane blind-issuance sessions, producing more than q distinct verifying ticket digests has probability at most the one-more-unforgeability advantage of either projected Blind-UOV lane, plus the dual-claw and ticket-hash collision terms.*

Proof. Map every completed issuance to the unique signed digest using Lemma 7.3. Project every complete ticket onto lane 0. More distinct verifying digests than completed sessions then give more valid lane-0 message/signature pairs than lane-0 responses, unless two payloads share a digest or Lemma 7.2 fails. Lane 1 remains mandatory for the cross-message binding target; it is not needed for this counting projection. $\square$

8 Candidate concrete instantiations

8.1 Augmented well-built-request compiler for Blind-UOV

The Blind-UOV framework already requires the user to prove that its masked request is well built. We strengthen that proved language instead of adding a separate policy proof. Let

$$R_{\text{wb}}^i(y_i; m, r_i, \rho_i) = 1$$

denote Equations (B3)–(B4) for lane i . The two native well-built-request languages are replaced by one augmented language:

$$R_{\text{aug}}(\text{pp}, \text{ctx}, \text{sid}, \text{rid}, y_0, y_1; M, (r_0, \rho_0), (r_1, \rho_1), k_{\text{hold}}, e) = 1 \iff R_{\text{issue}}(x; w) = 1,$$

where $\beta = (y_0, y_1)$ and Equation (I3) expands to $R_{\text{wb}}^0 \wedge R_{\text{wb}}^1$. The issuer executes the two ordinary Blind-UOV responses only after verifying a post-quantum SE-NIZK argument for R_{aug} .

Compiler interface. The integration requires a two-lane wrapper around the native verification hook:

$$\text{VerifyWellBuilt}(y_i, \pi_{1,i})_{i=0,1} \longrightarrow \Pi_{\text{issue}}.\text{Vfy}(\text{pp}, \text{ctx}, \text{sid}, \text{rid}, y_0, y_1, \pi_{\text{issue}}).$$

Each per-lane response and finalization algorithm remains unmodified. The spendable objects are two genuine Blind-UOV final signatures.

Concrete circuit boundary. The implementation SHOULD expose the following fixed interface to the proof compiler:

$$\begin{aligned} \text{pub}_{\text{aug}} &= (\text{ctx}, \text{sid}, \text{rid}, \text{tpk}, \text{bvk}, y_0, y_1), \\ \text{wit}_{\text{aug}} &= (\text{sn}, k_{\text{hold}}, e, (r_0, \rho_0), (r_1, \rho_1)). \end{aligned}$$

The circuit derives C , h , M , and m internally rather than accepting duplicate witness copies. Its constraint system is the conjunction

$$\mathcal{C}_{\text{aug}} = \mathcal{C}_{\text{shape}} \wedge \mathcal{C}_{\text{ticketHash}} \wedge \mathcal{C}_{\text{BUOVmask}} \wedge \mathcal{C}_{\text{holder}} \wedge \mathcal{C}_{\text{trace}},$$

where

$$\begin{aligned} \mathcal{C}_{\text{shape}} &: \text{TicketShape}(M, \text{ctx}) = 1, \\ \mathcal{C}_{\text{ticketHash}} &: m = H_{\text{ticket}}(\text{Encode}(M)), \\ \mathcal{C}_{\text{BUOVmask}} &: \bigwedge_{i=0}^1 [c_{r,i} = \text{CAP.Commit}_i(r_i; \rho_i) \wedge y_i = r_i + H_{\text{BUOV}}(\mu_i, c_{r,i})], \\ \mathcal{C}_{\text{holder}} &: h = H_{\text{hold}}(k_{\text{hold}}), \\ \mathcal{C}_{\text{trace}} &: R_{\text{Ntr}}(\text{tpk}, C, \text{ctx}, h; \text{rid}, \text{sn}, e) = 1. \end{aligned}$$

The proof statement includes sid even though it is absent from the ticket. Consequently, replaying the same proof in a new issuance session changes the statement and fails verification, while the final credential gains no session link tag.

Theorem 8.1 (Security inheritance of the augmented issuance proof). *Assume Π_{issue} is post-quantum zero knowledge and simulation-extractable for R_{aug} . Replacing the original well-built-request proof by π_{issue} preserves the projected Blind-UOV one-more property and parallel honest-protocol signer blindness, up to the SE-NIZK, dual-claw, and hash-collision terms already shown in Theorems 7.4 and 7.8.*

Proof. For one-more unforgeability, a forger against the augmented scheme is also a forger against the original Blind-UOV scheme: the reduction verifies the augmented proof itself and forwards each accepted y_i to its corresponding original signer instance. If that interface requires the native well-built proof, the reduction extracts (m, r_i, ρ_i) and generates it. Extraction failure is charged to SE-AoK. The augmented relation restricts both original well-built languages; each final signature format and verifier is unchanged.

For blindness, replace the augmented proof by a simulated proof. The remaining transcripts are ordinary Blind-UOV transcripts; a two-hybrid argument applies its blindness game once per lane. The public authenticated identity is session information but is absent from the final ticket; threshold-encryption privacy hides it inside $M.C$. Any attempt to make the same masked request represent a different ticket digest triggers Lemma 7.2, and any two payloads with one ticket digest trigger a collision in H_{ticket} . $\square$

Corrected concretization in version 1.6. The request and witness are now fixed:

$$\begin{aligned}\beta &= (y_0, y_1), & b &= ((r_0, \rho_0), (r_1, \rho_1)), \\ \mu_i &= \text{Encode}(\text{PQ-RBBC/BUOV-LANE}, i, H_{\text{ticket}}(\text{Encode}(M))), \\ c_{r,i} &= \text{CAP.Commit}_i(r_i; \rho_i), & y_i &= r_i + H_{\text{BUOV}}(\mu_i, c_{r,i}).\end{aligned}$$

The signer sees only the 64-byte pair (y_0, y_1) , not $c_{r,i}$ or m . Earlier drafts incorrectly exposed c_r and claimed that CAP binding plus hash collision resistance implied request uniqueness. That argument fails when an attacker chooses different commitments on both sides. Version 1.6 restores the paper ABI and states Assumption 3.2 explicitly. The removed ticket commitment c_M remains unnecessary: extraction supplies M , while ticket-hash collision resistance prevents a second payload under one digest.

Frozen Blind-UOV costing profile. Version 1.3 fixes the 31 October 2025 revision and its *Blind-UOV-Is*, *NIST-I*, *Shorter*, *TCitH* row. The paper gives $\bar{n}_r = 256$ bits, $\bar{n}_x = 640$ bits, a 66.6 KB public key, and a 3,772-byte final signature per lane. Its NIST-I/Shorter TCitH parameters are $\tau = 12$, ten 2^{11} -leaf trees plus two 2^{10} -leaf trees, $T_{\text{open}} = 111$, an explicit proof-of-work of 9 bits, and total proof-of-work 14.0 bits [1]. These values minimize the stored and online object; issuance is intentionally the expensive side of the trade-off.

Correct backend separation. Blind-UOV has two different proof objects. Its CAP commitment and final CAP proof are carried by the final blind signature. Before the signer responds, the user separately sends a NIZK π_1 proving that the masked request and CAP commitment are well formed. Our π_{issue} replaces and augments this π_1 ; it does *not* replace the final CAP proof. Consequently each 3,772-byte final signature remains unchanged. The dual profile carries two such signatures, while the ticket/trace blocks are shared inside one augmented issuance relation.

For costing, use the paper’s illustrative Shockwave/Anemoi row. For one native lane Blind-UOV-Is/Shorter is reported as approximately 16 million R1CS constraints, a 2.8 MB proof, 32 s proving, and 2.5 s verification. The dual baseline is therefore approximately 32 million constraints before adding the shared ticket/trace relation. The paper explicitly treats these as estimates and notes that the benchmarked implementations do not provide the required zero-knowledge mode. It also does not establish the simulation-extractability required by our concurrent GCCA proof. Therefore Shockwave/Anemoi is a *costing backend*, not yet the security-frozen Π_{issue} implementation [1].

The multivariate commitment $c = (\mu + F(r), G(r))$ and relation proofs of Bouillaguet et al. are a promising later optimization because they are algebraically aligned with UOV, but they add a newer MQ/bilinear-system assumption and are not the baseline for the first auditable construction [5].

8.2 Threshold Niederreiter for trace encryption

Code and plaintext profile. Let $\mathcal{G}$ be a binary Goppa code of length n , dimension k , and designed error weight t , with public systematic parity-check matrix $H_{\text{pub}} = (I_{n-k} \mid T)$. The selected decoder MUST guarantee $d_{\min}(\mathcal{G}) \geq 2t + 1$. Fix $\text{rid} \in \{0, 1\}^{256}$ and $\text{sn} \in \{0, 1\}^{128}$, and define the fixed 48-byte plaintext

$$\mu = \text{Encode}_{48}(\text{rid} \parallel \text{sn}).$$

The online ciphertext uses a weight- t error $e \in \mathbb{F}_2^n$ as its only encryption witness. Thus the abstract randomness ρ_{tr} is instantiated by e .

Hash DEM. Sample e uniformly from the weight- t vectors and compute

$$s = H_{\text{pub}} e^\top \in \mathbb{F}_2^{n-k}, \quad \text{wt}(e) = t, \quad (\text{N1})$$

$$\text{ad} = \text{Encode}(\text{ctx}, \text{sn}, h), \quad (\text{N2})$$

$$Z = \text{SHAKE256}(\text{PQ-RBBC}/\text{KDF}\|e\|s\|\text{ctx}, 640), \quad (\text{N3})$$

$$(K_{\text{mac}}, P) = Z, \quad |K_{\text{mac}}| = 32, \quad |P| = 48, \quad (\text{N4})$$

$$c_\mu = \mu \oplus P, \quad (\text{N5})$$

$$\gamma = \text{KMAC256}_{K_{\text{mac}}}(s\|c_\mu\|\text{ad}, 256; \text{custom} = \text{PQ-RBBC}/\text{TAG}), \quad (\text{N6})$$

$$C = (s, c_\mu, \gamma). \quad (\text{N7})$$

The output length in (N3) is in bits. SHAKE256 and KMAC256 are the standardized extendable-output and MAC functions of FIPS 202 and SP 800-185, respectively [10, 11]. The composition HDEM is nevertheless a new protocol profile, not a standardized AEAD and not yet a production claim. A standardized one-time AEAD may replace it later without changing the abstract R_{enc} interface.

Definition 8.2 (Concrete trace relation). *For $C = (s, c_\mu, \gamma)$, define*

$$R_{\text{Ntr}}(\text{tpk}, C, \text{ctx}, h; \text{rid}, \text{sn}, e) = 1$$

iff tpk selects H_{pub} and the fixed profile above, and Equations (N1)–(N7) all hold with $\mu = \text{Encode}_{48}(\text{rid}\|\text{sn})$. In this version,

$$\begin{aligned} R_{\text{enc}}(\text{tpk}, C, \text{Encode}(\text{ctx}, \text{sn}, h); \text{rid}\|\text{sn}, \rho_{\text{tr}}) &= 1 \\ \iff R_{\text{Ntr}}(\text{tpk}, C, \text{ctx}, h; \text{rid}, \text{sn}, e) &= 1, \quad \rho_{\text{tr}} = e. \end{aligned}$$

Hence $\mathcal{C}_{\text{trace}}$ is now the exact conjunction (N1)–(N7), not an unspecified “encrypt correctly” box.

Lemma 8.3 (Unique syndrome witness). *For a code with $d_{\min} \geq 2t + 1$, a syndrome s has at most one witness e of weight at most t satisfying $s = H_{\text{pub}} e^\top$.*

Proof. If e and e' were two such witnesses, then $d = e + e'$ would satisfy $H_{\text{pub}} d^\top = 0$ and $\text{wt}(d) \leq 2t$. If $e \neq e'$, d is a nonzero codeword below the minimum distance, a contradiction. Therefore $e = e'$. This is the standard unique-decoding argument used by Classic McEliece [8]. $\square$

Corollary 8.4 (Unique trace opening). *The concrete relation R_{Ntr} has information-theoretically unique plaintext openings.*

Proof. For a fixed C , Lemma 8.3 fixes e from s . Equations (N3)–(N4) then deterministically fix P and K_{mac} , and (N5) fixes the sole $\mu = c_\mu \oplus P$. The fixed-width encoding is injective, so it fixes (rid, sn) . Equation (N6) additionally rejects any inconsistent associated data. No computational assumption is needed for uniqueness. $\square$

Proposition 8.5 (Concrete trace correctness). *If threshold decoding reconstructs the unique error e for s , the combiner recomputes (N3)–(N6), returns $\mu = c_\mu \oplus P$, and accepts only if γ verifies and the decoded serial equals the visible sn . Consequently every honestly formed concrete trace ciphertext opens to exactly $\text{rid}\|\text{sn}$.*

Privacy assumption and proof boundary. Let t-OW-SD be the game in which an adversary receives H_{pub} , fewer than τ threshold shares, and $s = H_{\text{pub}} e^\top$ for uniform weight- t e , and must output e . We assume this advantage is negligible for the frozen distributed decoder.

Theorem 8.6 (Conditional concrete trace privacy). *Model SHAKE256 as a quantum random oracle, assume KMAC256 is a quantum-secure PRF, never reuse e , and let $\epsilon_{\text{O2H}}(q_H)$ denote the loss of the applicable one-way-to-hiding lemma for q_H quantum hash queries. Then for every fewer-than-threshold QPT distinguisher $\mathcal{A}$ there are QPT reductions $\mathcal{B}_1, \mathcal{B}_2$ such that*

$$\text{Adv}_{\text{Ntr}, \mathcal{A}}^{\text{t-IND-CPA}} \leq \text{Adv}_{\mathcal{G}, \mathcal{B}_1}^{\text{t-OW-SD}} + \epsilon_{\text{O2H}}(q_H) + \text{Adv}_{\text{KMAC256}, \mathcal{B}_2}^{\text{qPRF}}.$$

Proof. Starting from the real game, replace the 640-bit SHAKE256 value at the hidden input (e, s, ctx) by a uniform string. A distinguisher for this step either yields a threshold syndrome-decoding solver or is bounded by the stated one-way-to-hiding loss. Now P is an independent uniform 48-byte pad and K_{mac} is an independent uniform key. Replace KMAC256 under that key by a random function, paying the qPRF term. The value $c_\mu = \mu_b \oplus P$ and the tag are then independent of b , so the final advantage is zero. Summing the hybrid differences gives the bound. $\square$

A new accepted tuple (s, c'_μ, γ') with the same hidden derived key and different authenticated input gives a KMAC forgery, up to the same syndrome-decoding/QROM bad event. Thus the xor pad gives one-time privacy and KMAC256 gives one-time ciphertext integrity. Substituting Theorem 8.6 for the abstract τ -IND-CPA term in Theorem 7.11 yields protocol-level τ -GCCA, plus the SE-AoK, Blind-UOV, CAP, unique-opening, authorization, and hash terms already displayed there. This does *not* prove standard IND-CCA security for a bare ciphertext endpoint.

The threshold-Niederreiter work solves malformed-ciphertext attacks by attaching a straight-line-extractable validity proof to every public ciphertext; its authors report about 30.5 KB for a three-party ciphertext, with the proof dominating the syndrome [6, 7]. Here the same validity language $\{s : \exists e, \text{wt}(e) = t, s = He^\top\}$ is certified once inside π_{issue} . The online ticket retains only $C = (s, c_\mu, \gamma)$, and the raw distributed decoder remains unreachable.

Exact online byte counts. Because $|c_\mu| = 48$ and $|\gamma| = 32$, the trace ciphertext is exactly the syndrome length plus 80 bytes. Using the published Classic McEliece implementation sizes gives the following anchors [9]:

Parameter anchor	Public key	Syndrome	This C_{trace}
mceliece348864	261120 B	96 B	176 B
mceliece460896	524160 B	156 B	236 B
mceliece6688128	1044992 B	208 B	288 B
mceliece6960119	1047319 B	194 B	274 B
mceliece8192128	1357824 B	208 B	288 B

The aggressive working anchor is mceliece6688128: approximately 1.045 MB of cached public key buys a 288-byte trace ciphertext. The dual Blind-UOV-Is profile and exact 7,912-byte complete ticket are derived in Section 8.3; these are still research targets rather than implementation benchmarks.

Remark 8.7 (Mandatory 2026 cryptanalytic hold). *The anchor above is not frozen. A 10 August 2026 preprint gives a classical quasipolynomial-time distinguisher applying to all Classic McEliece NIST parameter sets and extends it to a heuristic quasipolynomial-time decryption attack; its concrete improvements are described as not yet practical [12]. That is not a demonstrated practical break, but it materially changes confidence. Deployment parameter selection is blocked until independent review quantifies this attack for the threshold construction.*

8.3 Reproducible five-block circuit audit

Counting model. The paper’s Anemoi profile uses an R1CS over $\mathbb{F}_{2^{129}}$ for NIST level I. Addition, xor, multiplication by public constants, fixed-width concatenation, and the public binary syndrome map are linear. Each Boolean AND or field multiplication contributes one nonlinear R1CS constraint. Following the Blind-UOV paper, one Keccak-f permutation costs $24 \cdot 1600 = 38,400$ constraints [1]. The audit counts only constraints added beyond two copies of the paper’s approximately 16-million-constraint Blind-UOV-Is/Shorter π_1 baseline.

Exact-weight gadget. The witness e has 6,688 bits. Pad it with 1,504 public zeros to 8,192 inputs and sum them using a fixed balanced binary adder tree. Combining two j -bit counters uses one half-adder and $j - 1$ full adders. Since xor is linear, the nonlinear cost is $1 + 2(j - 1) = 2j - 1$. The exact cost is therefore

$$\sum_{j=1}^{13} \frac{8192}{2^j} (2j - 1) = 24,547$$

AND constraints. Equating the final 14-bit count to 128 is linear. This gadget avoids the invalid shortcut $\sum e_i = 128$ in characteristic two, which would check only parity. Separately, e , sn , and k_{hold} require $6688 + 128 + 256 = 7,072$ bitness constraints.

Exact Keccak accounting. SHAKE256 has a 136-byte rate. With pad_{10^*1} , an L -byte message uses $\lfloor L/136 \rfloor + 1$ absorb permutations. The fixed encodings give:

Invocation	Absorbed bytes	Keccak-f calls	Nonlinear constraints
$H_{\text{ticket}}(M)$	$14 + 368 = 382$	3	115,200
$H_{\text{hold}}(k)$	$12 + 32 = 44$	1	38,400
Trace SHAKE KDF	$11 + 836 + 208 + 32 = 1087$	8	307,200
KMAC256	$136 + 136 + 336 + 3 = 611$	5	192,000
Total		17	652,800

For KMAC, the first 136-byte block is the cSHAKE function/customization prefix, the second is $\text{bytepad}(\text{encode_string}(K), 136)$, the authenticated data is $208 + 48 + 80 = 336$ bytes, and $\text{right_encode}(256)$ is three bytes. All squeezing fits in the first rate block and adds no permutation.

Five-block result. The incremental nonlinear counts are:

Block	Constraints	Contents
$\mathcal{C}_{\text{shape}}$	128	serial bitness; widths/wiring linear
$\mathcal{C}_{\text{ticketHash}}$	115,200	three SHAKE permutations
$\mathcal{C}_{\text{BUOVmask}}$	0	already in the paper baseline
$\mathcal{C}_{\text{holder}}$	38,656	256 bitness + one SHAKE permutation
$\mathcal{C}_{\text{trace}}$	530,435	e bitness, weight, KDF, and KMAC
Added relation	684,419	exact for the specified gadgets

The syndrome equation adds 1,664 linear output equalities but no nonlinear gate. The xor pad, associated-data construction, external/internal serial equality, and final constant comparisons are also linear.

Paper-anchored issuance estimate. Because the paper rounds each lane baseline to 16 million constraints, the combined number is itself approximate:

$$N_{\pi_{\text{issue}}} \approx 2(16,000,000) + 684,419 = 32,684,419,$$

or 2.0428 times the one-lane paper anchor. Applying the paper’s square-root proof-size scaling and linear prover scaling gives approximately 4.00 MB and 65.4 s. A conservative linear scaling of its verifier figure gives 5.11 s. These are extrapolations from Table 5 of the Blind-UOV paper, not local benchmarks. The paper does not provide enough information to derive peak prover memory, so memory remains a measurement obligation.

Proposition 8.8 (Monolithic-circuit feasibility decision). *Under the costing model above, adding the entire ticket and trace relation to the dual-lane well-built-request NIZK changes its baseline by about 2.14%. There is therefore no communication-based reason to split R_{aug} into linked proofs at this stage. A split is justified only if the selected simulation-extractable backend, implementation memory, or prover parallelism makes the monolithic circuit impractical.*

Exact online object under the frozen profile. For mceliece6688128, $|C| = 288$ bytes and

$$|M| = 32 + 16 + 32 + 288 = 368 \text{ bytes.}$$

Combining this with two published 3,772-byte Blind-UOV-Is/Shorter signatures gives

$$|T| = |M| + |\sigma_0| + |\sigma_1| = 7,912 \text{ bytes}$$

before transport framing. The cached public material is approximately 1,044,992 bytes for the trace key plus the paper’s rounded 66.6 KB Blind-UOV verification key, reused by both lanes. Relative to a 70 KB online credential, the frozen target is about 8.9 times smaller. The approximately 4.00 MB issuance NIZK is offline and is never forwarded to a satellite verifier.

Remark 8.9 (Audit reproducibility). *The companion script `pq_rbbc_circuit_audit.py` recomputes every integer above and emits the checked `pq_rbbc_circuit_audit_v1_3.json`. Exact gadget counts and paper-derived estimates are kept in separate JSON fields so a rounded source number cannot be mistaken for a benchmark.*

8.4 Executable reference relation and rejection tests

Version 1.6 extends the incremental relation in `pq_rbbc_reference.py`. The witness builder derives h , s , c_μ , γ , the canonical 368-byte payload, and its ticket digest from $(\text{rid}, \text{ctx}, \text{sn}, k_{\text{hold}}, e)$. The circuit builder then emits a stream of characteristic-two events of four kinds: linear wire definitions, linear equality assertions, Boolean multiplication gates, and bitness constraints $x(x + 1) = 0$. A backend can consume the same stream without retaining the whole circuit in memory. This version executes the stream and counts it, but does not yet flatten it into a proof system’s matrix format.

The symbolic Keccak-f implementation performs all 24 rounds. Its SHAKE256 output is checked against Python’s independent FIPS 202 implementation, and the SP 800-185 KMAC256 output is checked against the OpenSSL 3 KMAC provider [10, 11]. For the frozen honest vector the generated circuit reports:

Measured generator quantity	Value
Allocated wire identifiers	2,976,784
Linear definitions	2,281,323
Linear equality assertions	2,958
Nonlinear constraints	684,419
Public input bits	3,968
External Blind-UOV assertions	2
Keccak-f permutations	17
Failed assertions	0

Thus the executable result exactly reproduces every nonlinear block total in the preceding audit. The linear-definition figure is diagnostic IR size, not an additional nonlinear count: the costing model requires a backend to inline affine expressions. A backend that instead materializes each affine wire as an R1CS row must report that larger total explicitly. On the present reference environment, one honest streaming generation takes about six seconds. This is Python relation generation time, not NIZK proving time, and no prover-memory claim follows from it.

The regression suite accepts the honest witness and sends eight independent mutations through the complete circuit: wrong error weight, changed syndrome, changed masked identity, changed holder hash, changed KMAC tag, changed serial, changed blind request, and changed common context. All eight are rejected. The checked output is recorded in `pq_rbbc_reference_manifest_v1_6.json`.

Exact implementation boundary. The $H = (I \mid T)$ matrix in the reference vector is generated deterministically from a public seed and is only a rank-controlled test fixture; it is not a certified binary Goppa key. Similarly, `TestBlindUOVAdapter` checks message/request binding with a deterministic SHAKE construction, but is not a blind signature. It marks two native Blind-UOV π_1 /CAP boundaries whose nonlinear cost remains in the dual paper-anchored baseline. The 512 public y_0, y_1 bits are serialized as circuit inputs, while $m, r_i, \rho_i, c_{r,i}$ remain hidden. Consequently this implementation establishes that the incremental relation is internally consistent and executable; it does not establish that the combined native Blind-UOV proof is zero knowledge or simulation extractable.

8.5 Portable binary R1CS lowering

Version 1.6 uses the proof-backend exchange layer in `pq_rbbc_br1cs.py`. It consumes the same event stream as the counting sink and serializes three standard rank-one forms over $\mathbb{F}_2$:

$$\begin{aligned}
\text{MUL}(a, b, c) : \quad & a \cdot b = c, \\
\text{LINDEF}(c; x_1, \dots, x_j; k) : \quad & (c + x_1 + \dots + x_j + k) \cdot 1 = 0, \\
\text{ASSERT}(x_1, \dots, x_j; k) : \quad & (x_1 + \dots + x_j + k) \cdot 1 = 0.
\end{aligned}$$

Wire identifiers and arities use unsigned LEB128 encodings. A fixed 124-byte header commits to all row counts, the body length, and the SHA-256 digest of the body. Public and secret input declarations contain identifiers and visibility only; assignment values are not embedded in the circuit archive.

The resulting honest archive has the following exact dimensions:

Portable backend quantity	Value
Wires	2,976,784
Public input bits	3,968
Secret input bits	7,072
Nonlinear R1CS rows	684,419
Materialized affine-definition rows	2,281,323
Linear assertion rows	2,958
Total R1CS rows	2,968,700
Archive size	49,183,966 bytes

The 684,419 figure is therefore a *nonlinear-only* cost under the affine elimination model used in Section 8.3. The 2,968,700 figure is the portable total when every affine definition is materialized as its own rank-one row. They are not competing measurements. A concrete proof backend may recover the smaller representation by substituting affine expressions into multiplication rows, but that optimization requires its own soundness argument and resource benchmark.

Although the exchange format names $\mathbb{F}_2$ as its base relation, the same Boolean assignment and equations embed into $\mathbb{F}_{2^d}$: addition remains xor, and the input bitness equations prevent non-Boolean witnesses. This makes the archive compatible in principle with the $\mathbb{F}_{2^{129}}$ cost anchor, without claiming that a specific prover has been integrated.

An independent reader reconstructs and evaluates all 2,968,700 rows. It accepts the honest assignment, rejects a one-bit assignment mutation, and rejects a one-bit archive corruption through the body digest. Lowering a wrong-weight witness produces exactly the same 49,183,842-byte body and the same digest as the honest witness, showing that circuit structure is witness-independent; its assignment nevertheless fails 320 rows. Five end-to-end backend tests completed in approximately 77.9 seconds in the reference environment. The checked metadata is recorded in `pq_rbbc_br1cs_manifest_v1_6.json`.

Remaining external rows. The archive records two named external assertions for the native Blind-UOV π_1 /CAP request relations, but cannot evaluate them. This is deliberate and machine-visible: an archive with an unchecked external assertion is not a closed issuance proof. The next integration must replace that marker with the actual Blind-UOV constraints; deleting or treating either as automatically true is forbidden.

8.6 Local mixing as a future trace-encryption candidate

The core does not depend on elliptic curves, lattices, or codes. A future local-mixing encryption may replace threshold Niederreiter only if it supplies all four properties (T1)–(T4), an efficiently provable NP encryption relation, and a practical τ -party opening protocol. A small standalone ciphertext claim is insufficient by itself: threshold opening, unique opening, and extractable issuance binding are the missing system properties.

9 What has and has not been proved

Status	Result or obligation
Conditional abstract reductions	Correctness; dual proof/request/message binding; certified ciphertext validity; signature-gated decoder safety; issuer unlinkability against an honest-protocol curious issuer; trace soundness; one-more ticket unforgeability; and τ -GCCA trace privacy.
Assumed component property	Post-quantum SE-NIZK AoK; dual-lane cross-message request claw resistance; parallel Blind-UOV blindness and projected one-more unforgeability; threshold syndrome-decoding privacy with robust shares; QROM one-way-to-hiding; KMAC integrity; and authorization unforgeability.
Concretely instantiated	Blind-UOV-Is/Shorter costing profile; executable five-block augmented relation; exact Niederreiter–SHAKE–KMAC trace relation; information-theoretic unique opening; independently checked SHAKE/KMAC; full-circuit positive and negative tests; a checksummed portable R1CS archive with independent row evaluation; 684,419 incremental nonlinear rows; 2,968,700 materialized total rows; a 64-byte issuance request excluding proof; and a 7,912-byte online-ticket target.
Not yet instantiated	A post-quantum zero-knowledge and simulation-extractable implementation of Π_{issue} ; a structured-QROM reduction for Assumption 3.2; two native Blind-UOV circuit imports; proof-system-specific affine elimination; import of a real Goppa key; peak prover memory; robust threshold-share transcript; proof benchmarks; and a reviewed response to the August 2026 McEliece cryptanalysis.
Explicitly not claimed	Standard IND-CCA security of the bare ciphertext, security against a malicious issuer, metadata/timing anonymity without system controls, UC composability, side-channel resistance, satellite AKE security, or FLEO/handover security.

10 Implementation invariants

The following invariants are part of the cryptographic design and should be machine-tested or enforced by type/API boundaries.

- INV-1 There is no production function that accepts a bare trace ciphertext and returns a partial decryption.
- INV-2 The bytes hashed for the issuer signature are exactly the canonical bytes parsed by ticket verification and opening.
- INV-3 The issuer signs only after successful proof verification and records a completed session only after returning its blind response.
- INV-4 The blind request proven in Π_{issue} is byte-for-byte the request passed to the blind-signing implementation.

- INV-5 The public issuance request contains exactly (y_0, y_1) ; it contains neither m nor either $c_{r,i}$. The two lanes use independent masks and commitment randomness plus explicit lane-domain separation.
- INV-6 The opening authorization includes the ticket digest and evidence digest, and replay is statefully rejected.
- INV-7 The reconstructed serial number must equal the clear serial number.
- INV-8 Issuer keys and opening keys have independent generation ceremonies, storage, rotation, and compromise domains.
- INV-9 Epoch metadata is federation-fixed; the issuer cannot personalize it.
- INV-10 Every component and parameter set has an explicit post-quantum security claim at the chosen target category.
- INV-11 The parser accepts exactly the frozen 368-byte payload layout and rejects variable-length or alternate encodings before signature verification.
- INV-12 The issuance circuit enforces bitness of all 6,688 error coordinates and proves exact weight 128 with the specified counter gadget; a parity-only check is forbidden.
- INV-13 The Shockwave/Anemoi figures may be used for costing only until a backend meeting the stated zero-knowledge and simulation-extractability model is proved or substituted.
- INV-14 Production configuration rejects the deterministic parity-check matrix and test-only blind-signature adapter; neither fixture may cross the backend boundary.
- INV-15 Backend lowering must preserve every multiplication and equality event, eliminate affine wires soundly, and accept/reject the published reference and mutation vectors identically to the executable IR.
- INV-16 A serialized circuit is accepted only if its declared length, row counts, sequential wire discipline, and body digest all verify before proof generation or verification.
- INV-17 Any nonzero external-assertion count makes the archive incomplete for production issuance; both Blind-UOV markers must be replaced, never ignored.

11 Next proof step

The incremental relation, unique opening, fixed layout, circuit audit, executable witness builder, rejection tests, and portable R1CS lowering are now complete. The next step has two coupled proof obligations:

First prove Assumption 3.2 for the structured, domain-separated J^0, J^1 maps in the QROM, or replace the combiner with one that has such a theorem. In parallel, replace both external markers with the native Blind-UOV π_1 /CAP constraints and prove that both message inputs are exactly the 256-bit digest produced by $C_{\text{ticketHash}}$. Then select a concrete backend, prove any affine-elimination pass sound, and establish post-quantum zero knowledge and simulation extractability in the model required by the GCCA reduction.

The current count favors one monolithic proof; splitting is no longer the default. The Niederreiter deployment parameter MUST remain provisional under Remark 8.7. After backend qualification, the remaining cryptographic component is the robust threshold-share transcript. Satellite AKE integration remains out of scope until those two obligations close.

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